Corresponding Author Profile



Hussain Touhid Siddiquee · Researcher and Embedded Systems Engineer

Department of Electrical and Electronic Engineering, Leading University, Sylhet, Bangladesh

- ORCID: orcid.org/0009-0002-8804-6195
- Google Scholar: Citation profile
- Web of Science: Researcher profile (Publons)

About

Hussain Touhid Siddiquee is a researcher and embedded systems engineer in the Department of Electrical and Electronic Engineering (EEE) at Leading University in Sylhet, Bangladesh, where he completed his B.Sc. in Electrical and Electronic Engineering. His work brings together embedded systems, machine learning pipelines, battery health and failure prediction, solar energy systems, and full-stack software development. He contributes to academic research initiatives and to hardware and software integration projects.

Research interests

- Embedded systems
- Machine learning pipelines
- Battery health and failure prediction
- Solar energy systems
- Full-stack software development

1. Institutional Email

eee_0182220012131004@lus.ac.bd

2. Profile Page at the Institution's Website

Not available. Leading University does not maintain individual profile pages for students.

Alternative identifier: ORCID, orcid.org/0009-0002-8804-6195

3. Google Scholar Page

Google Scholar profile

4. Most Recent Published Papers

No papers have been formally published yet. Current manuscripts are listed below with their status as of October 2026.

1. **Helios-Artemis: Design and Simulation-Based Validation of a Dual-Microcontroller Predictive Solar MPPT Controller with On-Device LSTM Retraining for Sylhet Monsoon SHS Deployment in Bangladesh.** International Journal of Power Electronics and Drive Systems (IJPEDS). Status: accepted, pending publication (September 2026). Journal website
2. **Sampling and quantization limits of digitally controlled inverting buck-boost converters.** IJPEDS. Status: second round of revisions (October 2026). Journal website
3. **Machine learning in battery management systems: a structured scoping review and classification of 103 peer-reviewed studies, 2019–2026.** IEEE Transactions on Transportation Electrification. Status: first round of review (October 2026). Journal website
4. **Modeling electrical losses in CsPbI3 solar cells: stacked idealizations and a routine-quality envelope at 18.0%.** AIP Advances. Status: under review. Journal website

Education

B.Sc. Electrical and Electronic Engineering, Leading University, Sylhet, Bangladesh